

AVIATION COMMUNICATION INFRASTRUCTURE SECURITY

Aniruddha R. Karmarkar, Lockheed Martin, San Jose, CA

Abstract

In support of NextGen data communications applications, the FAA (Federal Aviation Authority) and the other ANSPs (Air Navigation Service Providers) plan to share the existing data radio currently used for supporting airline operations. Traditionally, ANSPs have operated all their communication in a relatively closed network environment. Air Traffic control data communications between the controller and the pilot; using existing Communication Service Provider (CSP) infrastructure over the existing data radio exposes the air traffic control message traffic to vulnerabilities in the network outside the ANSP domain. The use of shared radio and the ground infrastructure necessitates operation in a relatively open network environment where the ANSP network is interconnected with the CSP network which in turn is connected to airlines operations. Network hackers can probe for vulnerabilities in the network and launch a covert network attack at an appropriate time. While the air to ground communication security is subjected to collaborative standards evolution, there is a need to address the potential of distributed network attacks that may be launched from outside on the ground network. This paper explores potential vulnerabilities in the ground network, possible techniques to mitigate attacks, such as distributed denial of service attacks launched by hackers on the ground infrastructure outside the ANSP domain. The network attacks described in this paper are not restricted to data communications they are equally applicable to surveillance systems. The paper also explores the heightened possibility of penetrating the ANSP infrastructure firewall when network attacks are in progress.

Air to ground communication security is subjected to collaborative standards evolution. Most of the data in air-ground communication is tactical in nature and providing air-ground security is generally not considered a high priority because of myriad of issues that must be addressed for a successful implementation.

One of the important issues in this area is the need to address key management between the end users and systems in a highly mobile environment.

There are simplified approaches evolving for key management that should be carefully considered. In this paper we recommend an identity based key management as a possible solution to address these problems.

Background

Next Generation Air Transportation System (NextGen) will provide a more convenient and dependable air transportation system. As part of NextGen the FAA and other international ANSPs will use VDL Mode 2 radios to support ATC services using controller to pilot data communications. Data communications supplements the voice based communication system for ATC services. The CSP air-ground network and the VDL Mode 2 radio on the aircraft are also used for airlines operations. The ANSPs plan to share the existing CSP communication infrastructure for ATC traffic. The shared use of VDL Mode 2 radios opens up new cyber security vulnerabilities for ATC communications. The issue of network and cyber security using commercial infrastructure for data communications is a new paradigm for the ANSP's current relatively closed networks and systems.

The ANSPs will use the current voice based communication system as a backup system for data communications. Data communications based enterprise applications will enable trajectory based operations, resulting in increased use of automation for air traffic operations. Once data communications is widely deployed over next 10 to 15 years, the use of the backup voice network for air traffic operations will become infrequent.

The Data Communications necessitates operation in a relatively open network environment where the ANSP network is connected to the CSP networks which in turn are connected to airline operations. While air to ground communication security is subject to collaborative standards evolution, the potential of distributed network attacks

that may be launched on the CSP ground networks must be considered.

The consequent cyber security vulnerabilities in this environment are not only new but were not previously anticipated. The existing code of federal regulations does not adequately address cyber security vulnerabilities for open collaborative networks. Consequently, there are no existing policies, certification criteria or procedures that provide assurances that cyber security vulnerabilities will not cause unsafe flight conditions. Unmitigated, these vulnerabilities could have a detrimental effect on flight safety.

In this paper we provide a brief overview of the potential security implications for the ANSPs, we then describe the potential network security and cyber security issues, and then provide potential solutions for addressing those issues.

Data Communication Overview

In this section, we provide an overview of the proposed data communications network and outline some of the security considerations associated with the data communications network.

Figure 1 illustrates a high-level view of the data communications network showing the interconnections between the Airline Operations centers and the CSP network, and the ANSP.

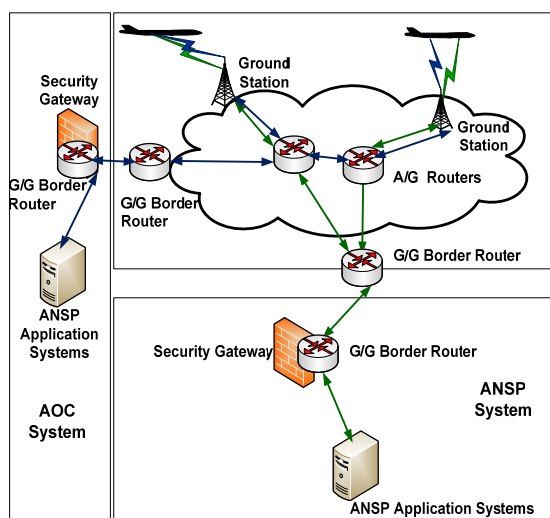


Figure 1. Data Communications Network High Level View

In general, we can make the following observations about the security considerations surrounding this network architecture:

- CSP network components are shared between best effort airline operations centers using AOC (Aeronautical Operational Control) services, and ANSPs mission-critical ATC traffic. Such disparate network components are typically not certified (from a security perspective) to the same degree; potentially allowing malicious actors or misconfiguration in the airline operations centers to affect ATC services.
- The network architecture has certain critical component and capacity bottlenecks. In particular, the maximum data rate supported by the VDL mode 2 air-ground networks is 31.5 kbps, which is smaller than the ground network circuits, which are typically, leased lines with aggregate bandwidths ranging from 56 Kbps-2.048 Mbps. This mismatch in bandwidth implies that a malicious or misconfigured end system may easily be able to flood a VGS subnet, resulting in a denial of service attack affecting aircraft serviced by that subnet.
- Average usage rates at a VGS station are typically an order of magnitude lower than the provisioned network circuits. A successful network attacker can further exploit the mismatch between average usage and available capacity by opportunistically opening up a small covert channel for ex-fil of the critical data. The ex-filled data can be further analyzed and possibly used to exploit vulnerabilities and gain unauthorized access and then infect network nodes and application systems within the CSP network and other adjacent networks with malware, and botnets.
- The CSP networks use commercial Internet Protocols (IP) for interconnection to airline operations centers. Commercial internet consists of a large number of interconnected networks operated as different autonomous systems (AS).

Border Gateway Protocol (BGP) is an Inter-domain Routing Protocol widely used for IP (Internet Protocol) networks. BGP is the IP stack equivalent of Inter-domain Routing Protocol (IDRP) protocol used for the ATN (Aeronautical Telecommunication Network) stack. Routing protocols, generally the Border Gateway Protocol (BGP), determine forwarding paths between end hosts in the enterprise. BGP update messages are exchanged between peer AS networks. These update messages provide information on available paths and end destinations reachable through each AS. By design, each AS trusts the information it receives via BGP update messages. Spoofing refers to the forgery of identity information (e.g. IP addresses in packet headers), typically by malicious attackers in order to avoid detection, or prevent source identification. Spoofed or incorrect BGP update information from any network can easily propagate through multiple ASes over multiple paths to corrupt routing tables and can affect network denials of service. The use of routing protocols such as IDRP within the context of data communications implies denial of service is a real possibility for ATC traffic.

Carefully considered usage of best-practice security recommendations can mitigate several of these threats and reduce the impact of compromised elements in commercial components such as the airline operations center.

Network and Cyber Security Context for Data Communication

With the widespread use of networking, information technologies, and networking and mobile communication technologies, malicious actors are continually seeking out cyber vulnerabilities in systems and gaining access to sensitive and protected data.

Cybersecurity issues transcend the protection of personal data or networks from hackers or even organized crime. Cyber warfare is a major national security issue. Protecting the security and freedom of our networks is as critical as protecting freedom of

the seas and space. It is difficult to establish evidence that proves beyond a reasonable doubt that a particular entity staged an attack. Not only is it difficult to identify and prove whether the attacker is sanctioned by a foreign government, it is also hard to distinguish between active direction by foreign officials and mere tolerance or lax enforcement. Consequently, accountability for cyberattacks is extremely difficult to determine. [1].

Recent cyber security incidents such as the widespread disclosure of highly classified diplomatic communication data by WikiLeaks, Stuxnet computer virus attack on Iran's Bushehr nuclear plant, the federal government Department of Energy (DOE) experiment to test the vulnerabilities of the US power grids, and the reported vulnerabilities of the surveillance systems such as ADS-B (Automatic Dependent Surveillance-Broadcast) ([2] indicates that the use of data communications for ATC communication should address cyber security vulnerabilities.

Adversaries have goals and objectives. They have requisite resources and they have risk tolerance. For each CSP, ANSP, and AOC system it will be worthwhile to analyze the type of adversaries - how much of their resources are they willing to expend and how much risk will they take with respect to being detected or discovered?

It will be worthwhile to consider system vulnerabilities of ANSP, CSP, and AOC systems in the context of adversaries, their motives, and the resources at their disposal. These vulnerabilities must be further analyzed further to determine what resources are required to effectively exploit the vulnerability? What is the probability of detection? How long does it take to exploit the vulnerability? The possible impact if the vulnerability is exploited successfully to attain an adversary objective.

Risk is the potential for realization of any threat and the associated impact. Mitigations are mechanisms to prevent realization of threats. Considerations for mitigation strategy should consider how much does it cost to implement? Which threats does mitigation prevent or mitigate? Is there a quantitative measure to determine the degree of effectiveness? The end goal is not to mitigate all the known threats. Objective is to minimize risk within reasonable cost.

Misconfigurations, Misdirections and Spoofing

The YouTube Pakistan incident from 2008 highlighted several important cyber security issues.

All routing protocols have inherent design weaknesses because of the implicit trust each node places in updates it receives and because of the transitive nature of that trust. Errors and misconfigurations can and do occur. They are the largest source of issues in network carriers.

Malicious adversaries have additional options that are more subtle than that used in the YouTube incident. Spoofing addresses, altering routing tables, DNS cache poisoning are some of the techniques commonly used for misdirection, and spoofing.

Distributed Denial of Service

A denial-of-service, or DoS, attack occurs when someone directs a large number of requests to a target machine so quickly that the machine can't respond, and the site becomes inaccessible. A distributed denial-of-service, or DDoS, attack occurs when hundreds or thousands of compromised computers are enlisted to launch a denial-of-service attack.

The Estonian Distributed Denial of service attack from April 2007 and the earlier TitanRain attack in 2003 on US systems are the most well-known targeted Denial of service attacks.

These cyber attacks were able to shut down some sites for a long time. Network infrastructure was effectively unavailable for a long time. The events proved extremely time consuming, expensive to combat and indicative of weaknesses in cyber security.

The interconnection of international CSP and airline commercial networks to the data communications network exponentially increases the risk of potential DDoS attacks being launched against ATC infrastructure by compromised computers in the commercial domain.

Targeted Malware and Botnets

In many organizations it is believed that the use of highly customized proprietary software application implementation prevents the systems from attacks normally observed in the public domain internet. The Stuxnet computer virus attack demonstrated that this is a misplaced belief. Stuxnet is of particular interest

in the context of the data communications program and NextGen applications. Stuxnet is part virus, part a botnet.

Botnets are a collection of compromised computer systems controlled by adversaries. Attacker's command and control typically uses multiple layers to obscure master server identity. Each layer often includes fault tolerance capability to provide resiliency. Different layers and elements of attacker command and control are generally geographically distributed. Nodes in the command and control structure often rotate addresses, roles, domain names, etc. to hide their communications patterns. Adversaries use various techniques to avoid detection and attribution.

Stuxnet attack on Iran's Bushehr nuclear power plant highlighted the need for sophisticated defenses to guard against targeted malware and botnet type attacks in data communications networks.

Recommended Security Approach for Data Communication

In this section, we delineate the broad, best practice based approach to securing the data communications network against various threats. We utilize a "defense in depth" approach for addressing cyber security issues for the data communications.

Defense in Depth

A defendable computing and network environment use a Defense in-Depth strategy, best-practice techniques, Government-guidance and industry-standards compliance to gain an accurate view of security controls and design solution to identify and respond to genuine threats. Short-falls in a single-layer Information Assurance approach results in attack vulnerabilities in the event of malfunctions, misconfigurations, or security compromises. Defense-in-Depth mitigates exposure from new threats because an attack that successfully exploits a given vulnerability must still overcome additional security mechanisms in order to compromise or corrupt the system.

Good cyber security architecture utilizes seven unique Defense-in-Depth layers, comprising of a mixture of technology, people, and operations mechanisms that reduce the overall exposure of the

system. At least four layers of defense are required at every element and two layers of defense are required at every link of the network.

Defense-in-Depth architecture is built using a seven-layer approach (see Figure 2) and is implemented in infrastructure as:

1. **Data:** Risks at the data layer arise from vulnerabilities attackers could exploit to gain access to configuration data, organization's sensitive data, or any data that is unique to a device the organization uses. This layer is primarily the responsibility of each security domain peering with another security domain.

2. **Application:** Risks at the application layer arise from vulnerabilities an attacker could potentially exploit to running applications.

3. **Host:** Risks at this layer arise from attackers exploiting vulnerabilities in the services that the host or device offers.

4. **Internal Network:** Risks to internal networks largely concern the sensitive data transmitted via networks.

5. **Perimeter Network:** Risks associated with the perimeter network layer arise from an attacker gaining access to boundary connections (such as internal peers and external domains).

6. **Physical Security:** Risks at the physical layer arise from an attacker gaining physical access to the asset itself.

7. **Policies, Procedures and Awareness:** Surrounding the entire security model is a policies, procedures, and awareness layer that support the requirement for each level.

Defense in depth approach extends to all aspects of the data communications network including physical security of sites, all the installed software, deployed hardware elements, protection of the network message traffic, routing protocols and network management protocols. A comprehensive analysis of the vulnerabilities, potential threats arising out of the vulnerabilities, the effective risk of realization of those threats and the potential system impact should be studied. Use of well established operational policies, procedures, and network operator training minimizes the security risks to an acceptable level. The end goal is to protect the data

communications network against known network vulnerabilities and establish mechanisms to limit the impact to the network from zero day attacks.

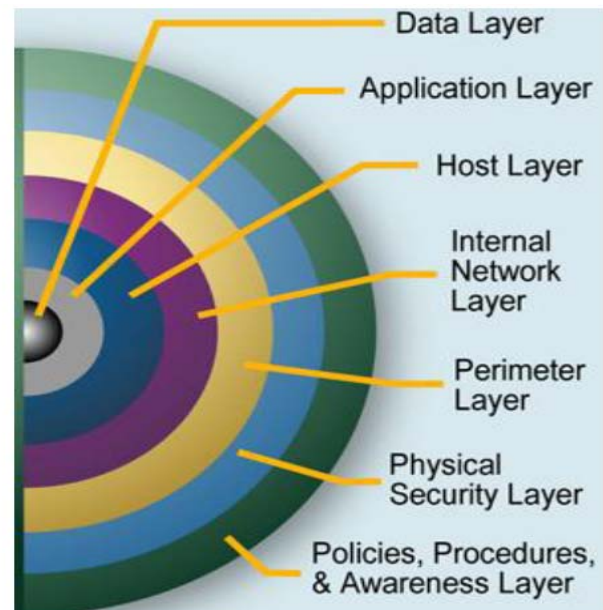


Figure 2. 7 Layers of Defense-in-Depth

Carefully considered usage of best-practice security recommendations can mitigate threats and reduce the impact of compromised elements in commercial components such as the airline operations center. We suggest the following measures to be deployed in the data communications network environment:

Metering and Rate-Limiting

Metering and rate-limiting are effective tactics for safeguarding against denial of service attacks, in particular excessive AOC traffic against a particular VGS station or subnet[3]. Metering involves limiting the number of messages per second initiated by specific end-systems based on their normal usage profiles, i.e. source-based metering. Further, limits on normal AOC usage per VDL subnet can be enforced based on historical observation or traffic demand prediction. This constitutes destination-based metering, i.e. controlling the amount of traffic permitted per destination VDL subnet.

Pairwise Communication Using Access Lists

Access lists can be used to ensure that pairwise communications can only take place between authorized entities, e.g. filtering out source-destination address pairs that are not permitted; thus

ensuring that ATC directives do not originate from the airline operations network. In conjunction, anti-source spoofing measures [4] can be used to ensure that sources do not misrepresent themselves by masquerading as other entities.

Preventing Misconfigurations

Human error is a major cause of misconfiguration, network service disruption, and degradation. Automated routing policy management can significantly reduce configuration errors and mitigate service disruption risk. Routing policy management is best performed as part of the network management system and consists of route registration, configuration generation, and configuration deployment.

Route Registration

During the provisioning process and subsequently when mission or network conditions require, the ANSPs AOCs and CSPs exchange lists of known, valid and invalid prefixes that might originate from the ANSP, AOC and CSPs for each peering relationship. This information helps define transit (CSP) network import policies and user (AOC and ANSP) network export policies. Other similar prefix lists help define transit (CSP) network export and user (AOC and ANSP) network import policies. The network administrators store import and export policies in a “database”, the route registry, using Routing Policy Specification Language (RPSL).

Configuration Generation and Deployment

After changes to the route registry, the network administrators initiate a configuration generator that reads the route registry and generates device specific routing configuration for each border router in the network. After generating configurations, the network management system sends device specific configurations to each border router for deployment into operations using existing configuration interfaces.

Initially created by Merit Network, Inc. as the Routing Assets Database (RADB), Internet Route Registry (IRR) [5, 6] provides a route registry and configuration generation capability. The IRR database stores routing policies using RPSL [7, 8] a human friendly, device independent language for routing policy. The configuration generator, called rtConfig, converts RPSL to device specific configurations. IRR is freely available, community-

tested by large commercial provider networks, and includes multi-vendor support.

In commercial use, IRR is best known as a public registry of routing information for networks. Organizations that operate networks (ISPs, universities, business enterprises) publish their routing policy and route announcements in IRR to facilitate the operation of the Internet. They use information in IRR to troubleshoot routing problems, generate access lists, and automatically configure backbone routers.

Benefits of Route Registry

Minimization of Human Error

Device specific configurations are automatically generated based on objects and policies in the IRR. Manual effort is minimized.

Improved Cyber Security

When users register routes or prefixes, there is greater confidence in and better definition of Martian routes. Attempts to spoof routes in the registry are blocked at each border router based on a priori configuration thereby enhancing routing security.

Use of route registry reduces denial of service attack vectors and dependence on dynamic routing protocols without trust vectors. This could be a particular problem for ATN communications, due to the use of the IDRP routing.

Route registry data should be protected from cyber attacks. A cyber attack against route registry can result in crippling the network and thus affecting the air traffic. Use of Byzantine algorithms [9, 10] for Registry distribution will ensure that compromise of registry database will not affect system availability.

Byzantine algorithms are a class of fault tolerant algorithms. The algorithms were proposed more than 25 years back by Leslie Lamport in his pioneering work on Byzantine General's Problem. The algorithms protect distributed systems against Byzantine failures. Byzantine failures occur when a system fails in an arbitrary manner i.e. provides corrupted information or corrupted local state. Signatures for Byzantine failures due to natural system failures and intentional malicious attacks are often similar making it harder to detect the original cause of the problem. This makes pinpointing the issue of attribution of cyber attack to an adversary very difficult. Byzantine failures are a major topic of research on cyber security.

Preventing Malware and Botnets

Often attackers initially probe for vulnerabilities at the network layer. Once they identify the vulnerabilities they launch a covert network attack at an appropriate time. A successful network attack often provides the attacker additional means to analyze the network data. The data may be ex-filled if a covert channel is available, otherwise other vulnerabilities can be studied in situ. The attacker can then introduce malware and botnets on the victim's network nodes and application systems.

Encryption and Key Management

The use of end-to-end encryption and authentication schemes at application layer provide good data integrity and can mitigate concerns over the validity of ATC (Air Traffic Control) directives between controller and pilot. While this is the most desirable end-state option for mitigation of malware and botnets, the use of such schemes requires standards evolution and mature implementation. Commercial strength crypto devices supporting Advanced Encryption Standard (AES) are available for cipher blocks of 128 bits/192 bits/256 bits. Key sizes of 128 bits (AES-128), 192 bits (AES-192), or 256 bits (AES-256) are readily available. They may be adequate for Data encryption. The mechanism for key distribution between the pilot and the controller will have to be addressed.

A X.509- like PKI infrastructure will be required for ensuring the identity of the two end users. VDL (VHF Data Link) Mode 2 uses a narrowband channel of 25 KHz. Pilot to controller communications over a Carrier Sense Multiple Access (CSMA)/ Collision Detect (CD) Media Access Control (MAC) layer, and the movement of the aircraft across VGS (VHF Ground Station) boundaries requiring frequent link layer handover mechanisms in a crowded air space will require frequent synchronization between the encryption devices at the controller and the pilot. This means establishing the encrypted channel between the pilot and the controller will lead to longer handover times. The same procedure will have to be repeated if the air/ground communication drops a link layer frame over the air. This may make the air/ground communication secure but will not enhance the overall operational safety of the system. There are simplified approaches evolving for synchronization that should be carefully considered.

Table 1. Recommended Security Measures

<i>Threat</i>	<i>Recommended measure</i>
Physical attack on network infrastructure	Physical Perimeter security Personnel access to facilities Network redundancy – network path and network equipment location diversity to ensure continuity of operation.
Compromise of hardware encryption (crypto) elements	FIPS 140-2 Level 1 or higher certification.
Compromise of software	Appropriate identification, authorization, authentication of user. Appropriate access controls and maintaining audit trail and chain of custody of information for people and systems.
Network attack via CSP non-ANSP interfaces leading to Denial of Service, and other attacks	Rate limiting through metering • Source-based metering. • Destination-based metering • Access lists to ensure pair-wise communications can only take place between authorized entities. • Use of message guards at the appropriate boundaries.
Misconfiguration resulting in attack back-doors -	Registration During the configuration process and subsequently when mission or network conditions require, the ANSP, AOC, and CSPs exchange lists of known, valid addresses.
Known attack vectors in the protocol stack, interfaces, software etc.	Hardening the appropriate protocol stack, interfaces, software as per established policies, procedures guiding security
Zero day (Not currently known) attacks	Mechanisms to isolate the infected infrastructure and operate through good network components. Use of techniques such as Byzantine algorithms to detect state compromises.

Key Management

A Key management system enables data encryption of data at the sender, decryption of data at the receiver. Every key management system requires a key server. A key server and underlying infrastructure authenticate the users and deliver the encryption keys to the sender, and deliver decryption keys to the receiver, enable joint management of keys with partners, be able to deliver keys to infrastructure components, be able to recover keys and able to scale.

Symmetric key management is the oldest mechanism. In this scheme same key is used to encrypt and decrypt information, the key manager generates a new key for every message at the sender's request. The key is stored along with the list of receivers. When the receiver authenticates the key is retrieved and delivered to the receiver if the receiver is authorized to decrypt the information from the sender.

Public Key Infrastructure (PKI) Key Management uses asymmetric keys. These systems use a different key to encrypt data than the one they use to decrypt data. The famous Diffie-Hellman and RSA algorithms are the best-known examples of public key algorithms. A recipient generates a pair of keys: one public key and one private key. To encrypt data, the sender generates a bulk encryption key, encrypts the bulk key with the recipient's public key, and sends the data along with the newly encrypted bulk key. The recipient gets the data, decrypts the bulk key with his private key, and then uses that key to decrypt the data.

PKI has two significant disadvantages: 1) generating the private key at the recipient makes key recovery difficult; and 2) the sender must locate a public key for each recipient and verify its validity. Since the recipients generate these keys themselves, the server may not be able to supply keys for all recipients. This problem is solved using certificates. Certificates are authenticated data structures that tie a receiver to a public key. Authenticated certificates can be stored on distributed, un-trusted directories, rather than the key server. This splits the key management server into a public facing directory and a certifying authority (CA). The CA is the only trusted component. The un-trusted directory handles most of the transactions in the system, enabling scalability.

Certificate-based systems cannot locate a certificate for a specific user, making them unable to encrypt the data.

Identity Based Key Management

Adi Shamir, one of the pioneers of public key cryptography, proposed a new type of public key algorithm in 1984. Shamir proposed mathematically generating the receiver's public key from his/her identity, and then having the key server calculate the required private key. This system, called an "Identity-Based Encryption" (IBE) algorithm, would remove the need for public key queries or certificates.

The Boneh/Franklin's pairing-based encryption scheme and Cocks's encryption scheme based on quadratic residues both solved the IBE problem in 2001.

Using IBE radically simplifies key management because the sender does not need to contact the key server to get an encryption key. Instead, the encryption key is mathematically derived from the receiver's identity. The receiver must only contact the key server once to authenticate and get the required decryption key. The key server is able to construct the receiver's decryption key mathematically.

In the context of the data communications between the controller and the pilot IBE scheme provides a special promise. The communication is always initiated by the pilot to the controller. The sender pilot can specify receivers (ANSP processing centers) and encrypt the message without a need to receive an encrypt key from the directory or the key manager. A receiver requires a single authentication to get the decrypt key. This scheme is highly scalable and robust. With this scheme the overhead on the air-ground communication can be minimized.

We recognize that a lot of work is still needed in this area before it will be practical to implement application layer secure encrypted channel between the pilot and the controller

In the meanwhile we recommend a more practical approach, a hop by hop encryption of links within the CSP infrastructure and on all external links. Use of relatively cheap commercial encryptor/decryptor devices supporting relatively low data rates in the ground segment may not be as expensive as installing and coordinating deployment

of commercial grade encryptor/decryptor devices in the avionics at a low overall cost.

In addition to installing the crypto devices we recommend installing message guards performing deep packet inspection at boundary between the ANSP and the CSP.

Conclusion

Data communications is a key enabler of NextGen applications. The use of data communications between controllers and pilots and over a shared RF network between airline operations and air traffic messages introduces new threat models that need to be carefully considered. In this paper, we have described applicable threats and outlined a best-practice based security approach that can be leveraged to mitigate or minimize the impact of potential threats and breaches to the data communications network. We believe that more detailed study is required in this critical area, to ensure smooth, secure transition to data communications as the primary channel between controllers and pilots.

References

- [1] Michael Chertoff, Former DHS Chief, March 11, 2010 "Cyber Shockwave Exposed Missing Links in US Security" in Federal Computer Week.
- [2] Paul Marks, September 2011, Air Traffic System Vulnerable To Cyber Attack, New Scientist
- [3] Aniruddha Karmarkar, Ravi Vaidyanathan, Oct.2011, "Security Implications for Data Comm" ATCA
- [4] Ravi Vaidyanathan et al, "On the use of BGP AS Numbers to Detect Spoofing", Globecom WPS 2010
- [5] Internet Routing Registry, Merit Network Inc., <http://www.irr.net/>
- [6] Internet Routing Registry Daemon, Merit Network Inc., <http://www.ird.net/>
- [7] C. Alaettinoglu et. al., June 1999, IETF RFC 2622, "Routing Policy Specification Language (RPSL)",
- [8] Blunk et. al., March 2005, IETF RFC 4012, "Routing Policy Specification Language next generation (RPSLNg)",
- [9] M. Pease, R. Shostak, L. Lamport *ACM* April 1980 "Reaching agreement in the presence of faults".
- [10] L. Lamport, R. Shostak, M. Pease, *ACM* July 1982, The Byzantine Generals Problem.

Acknowledgements

The author would like to acknowledge Fred Bay, Steve Hansen for their insightful comments.

*2012 Integrated Communications Navigation
and Surveillance (ICNS) Conference
April 24-26, 2012*